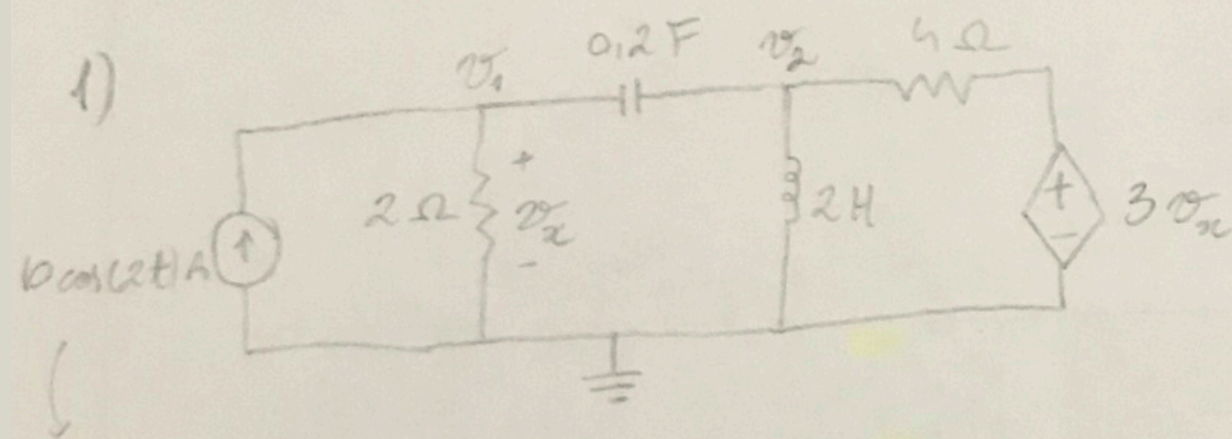
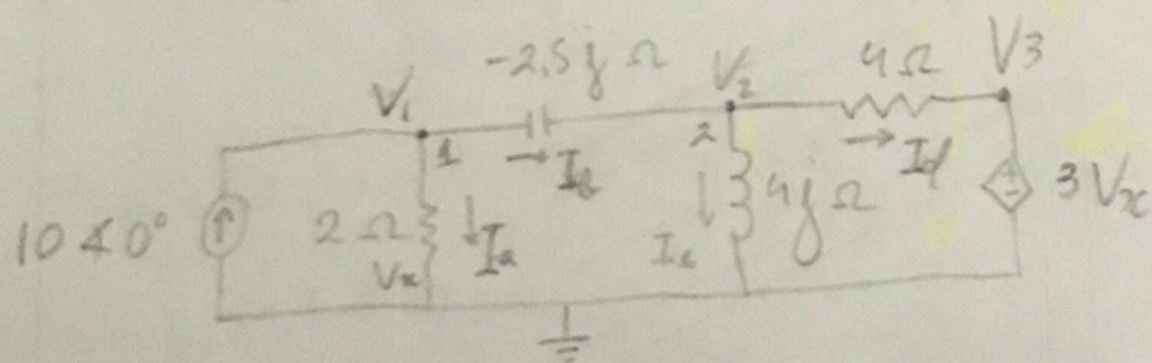


1)



$$\omega = 2 \Rightarrow \begin{cases} Z_C = \frac{1}{j\omega C} = \frac{1}{0.4j} = -2.5j \Omega \\ Z_L = j\omega L = 4j \Omega \end{cases}$$



Análise
nodal:

nó 1: $10\angle 0^\circ = I_a + I_b$

$$10 = \frac{V_1}{2} + \frac{V_1 - V_2}{-2.5j} = \frac{5V_1 + 4(V_1 - V_2)j}{10}$$

nó 2: $\frac{V_1 - V_2}{-2.5j} = \frac{V_2}{4j} + \frac{V_2 - V_3}{4}$

$$\begin{cases} V_1 = v_x \\ V_3 = 3v_x \end{cases} \Rightarrow V_3 = 3V_1$$

$$\Rightarrow \frac{8(V_1 - V_2)j}{20} = -\frac{5V_2 j}{20} + \frac{5V_2 - 15V_1}{20}$$

$$V_1(15 + 8j) = V_2(5 + 3j) \Rightarrow V_2 = \frac{V_1(15 + 8j)}{5 + 3j}$$

Substituindo na eq. do nó 1:

$$\frac{5V_1 + 4(V_1 - V_2)j}{10} = 10 \rightarrow V_1(5 + 4j) - 4V_2j = 100$$

$$\hookrightarrow V_1(5 + 4j) - 4j \cdot \left[\frac{V_1(15 + 8j)}{5 + 3j} \right] = 100$$

$$V_1(5 + 4j) + \frac{V_1(32 - 60j)}{5 + 3j} = 100$$

$$V_1(5 + 4j) + \frac{V_1(-20 - 396j)}{39} = 100$$

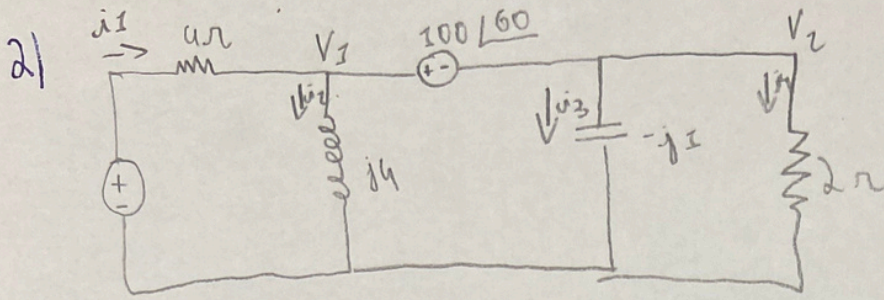
$$V_1(150 - 260j) = 3900 \Rightarrow V_1 = \frac{3900}{150 - 260j} = \frac{3900}{\sqrt{150^2 + 260^2}} \angle -(-\arctan(\frac{260}{150}))$$

$$\therefore V_1 = 11,327 \angle 60,018^\circ$$

$$\hookrightarrow V_2 = \frac{V_1(99 - 5j)}{39} = 11,327 \cdot \frac{99,126}{39} \angle 60,018^\circ + (-\arctan(\frac{5}{99}))$$
$$= 33,024 \angle 57,127^\circ$$

$$\Leftrightarrow v_1(t) = 11,327 \cos(2t + 60,02^\circ)$$

$$v_2(t) = 33,024 \cos(2t + 57,13^\circ)$$



$$i_1 = i_2 + i_3 + i_4 \approx$$

$$\Rightarrow \frac{75 - V_1}{4} = \frac{V_1}{j4} + \frac{V_2}{-j1} + \frac{V_2}{2}$$

$$\frac{V_2}{-j1} - \frac{1}{1} = \frac{jV_2}{j^2} \Rightarrow j = \sqrt{-1} \therefore j^2 = -1 \Rightarrow \frac{jV_2}{-(-1)} = jV_2; \frac{V_1}{j4} - \frac{j}{-j} = \frac{-jV_1}{-j^2 4} = -\frac{jV_1}{4}$$

$$\Rightarrow \frac{75 - V_1}{4} + \frac{j}{4} V_1 - jV_2 - \frac{V_2}{2} = 0 \xrightarrow{\star 4} 75 - V_1 + jV_1 - 4jV_2 - 2V_2 = 0 \Rightarrow$$

$$\Rightarrow (-V_1 + jV_1) + (-4jV_2 - 2V_2) = -75 \Rightarrow V_1(1 - j) + V_2(4j + 2) = 75$$

$$V_1 - V_2 = 100 \angle 60^\circ$$

$$V_1 = V_2 + 100 \angle 60^\circ = V_2 + [50 + j86,6025]; \text{ Vendo}$$

$$\text{Substituyendo: } [V_2 + (50 + j86,6025)] \cdot (1 - j) + V_2(2 + 4j) = 75 \Rightarrow$$

$$\Rightarrow (1 - j) \cdot (50 + j86,6025) + V_2[(1 - j) + (2 + 4j)] = (1 - j) \cdot (50 + j86,6025) + (3 + 3j)V_2 = 75$$

$$\star 50 + j86,6025 - j50 - j^2 86,6025 = 50 + j86,6025 - j50 + 86,6025 = 136,6025 + j36,6025$$

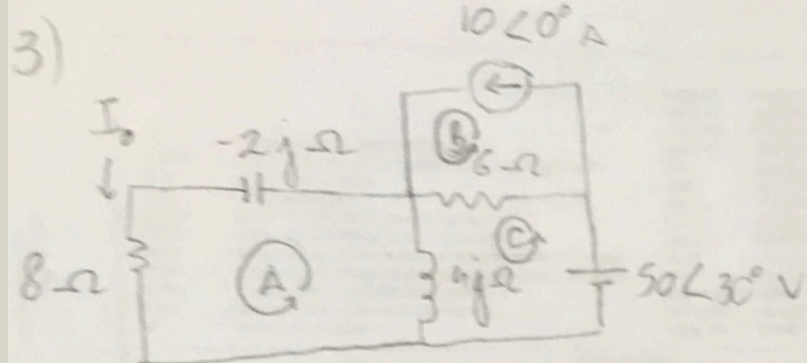
$$\Rightarrow (3 + 3j)V_2 + (136,6025 + j36,6025) = 75 \Rightarrow (3 + 3j)V_2 = 75 - 136,6025 - j36,6025 = -61,6025 - j36,6025$$

$$V_2 = \frac{-61,6025 - j36,6025}{3 + 3j} = \frac{-71,66 \angle 30,47^\circ - 180}{4,2426 \angle 45} = -16,88 \angle -134,29^\circ = 16,88 \angle -134,29^\circ + 360 =$$

$$= 16,88 \angle 165,71^\circ = V_2; V_1 = V_2 + 100 \angle 60^\circ = (-16,3674 + j4,16667) + (50 + j86,6025)$$

$$V_1 = 33,6325 + j90,7682 = 96,80 \angle 69,67^\circ; V_2 = 16,88 \angle 165,71^\circ //$$

$$V_1 = 33,6325 + j90,7682 = 96,80 \angle 69,67^\circ; V_2 = 16,88 \angle 165,71^\circ$$



Malha A: $8 \cdot I_A - 2j I_A + 4j(I_A - I_C) = 0$

$$I_A(8 + 2j) - 4j I_C = 0$$

$$I_C = \frac{4 + j}{2j} \cdot I_A = \frac{1 - 4j}{2} I_A$$

Malha C: $-50 \angle 30^\circ + 6(I_C - I_B) + 4j(I_C - I_A) = 0$

$$I_B = 10 \text{ A}$$

$$-50 \angle 30^\circ = -50 \left(\frac{\sqrt{3}}{2} + j \frac{1}{2} \right) = -25\sqrt{3} - 25j$$

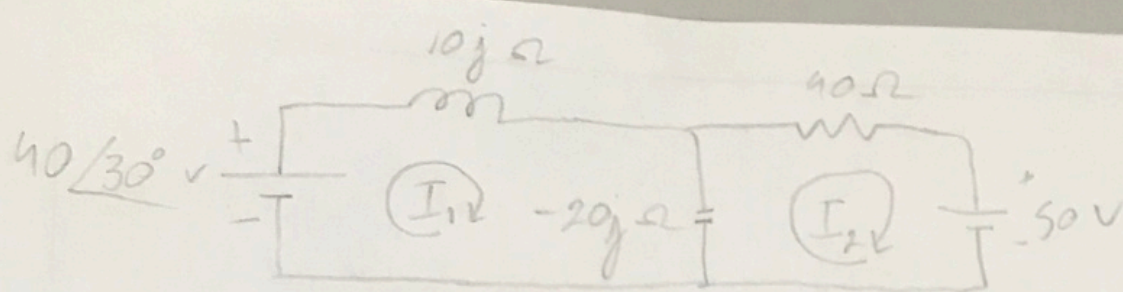
$$\rightarrow -25\sqrt{3} - 25j + 6(I_C - 10) + 4j(I_C - I_A) = 0$$

$$\Rightarrow I_C(6 + 4j) - 4j I_A = 60 + 25\sqrt{3} + 25j$$

$$\left(\frac{1 - 4j}{2} \cdot I_A \right) (6 + 4j) - 4j I_A = 60 + 25\sqrt{3} + 25j$$

$$(11 - 14j) I_A = 60 + 25\sqrt{3} + 25j$$

$$I_A = \frac{60 + 25\sqrt{3} + 25j}{11 - 14j} = \frac{106,28 \angle 13,605^\circ}{17,804 \angle -51,843^\circ} = 5,969 \angle 65,448^\circ$$



malha 1: $-10j I_1 + 20j I_2 = 40(\frac{\sqrt{3}}{2} + \frac{1}{2}j)$

Isolando I_2 , $I_2 = \frac{2}{j}(\frac{\sqrt{3}}{2} + \frac{1}{2}j) + \frac{I_1}{2} = -\sqrt{3}j + 1 + \frac{I_1}{2}$

malha 2: $(40 - 20j) I_2 + 20j I_1 = -50$

$\Rightarrow (40 - 20j)(-\sqrt{3}j + 1 + \frac{I_1}{2}) + 20j I_1 = -50$

$= -40\sqrt{3}j - 20\sqrt{3} + 40 - 20j + 20j I_1 - 10j I_1 + 20j I_1 = -50$

$\Rightarrow 10 I_1 (2 + j) = 40 - \sqrt{3}j + 20\sqrt{3} - 40 + 20j$

$I_1 = \frac{-55,359 + 89,282j}{20 + 10j} = \frac{105,05 \angle 121,80^\circ}{22,361 \angle 26,565^\circ}$

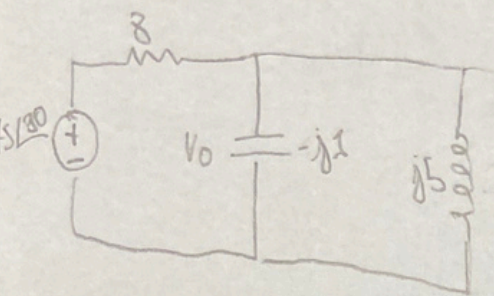
$I_1 = 4,968 \angle 95,235^\circ$

$I_2 = -\sqrt{3}j + 1 + \frac{-55,359 + 89,282j}{40 + 20j} = \frac{-40\sqrt{3}j + 40 + 20\sqrt{3} + 20j - 55,359 + 89,282j}{40 + 20j}$

$I_2 = \frac{19,282 + 40j}{40 + 20j} = \frac{44,405 \angle 69,264^\circ}{44,721 \angle 26,565^\circ}$

$\Rightarrow I_2 = 0,9929 \angle 37,699^\circ$

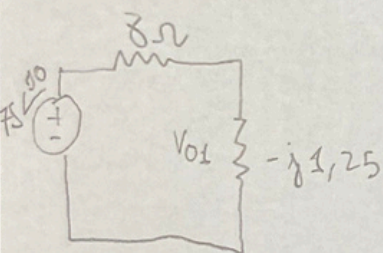
5) $V_1 = 75 \angle 90^\circ$; $I_1 = 6 \angle 0^\circ$; $Z_1 = 8 \Omega$; $Z_2 = \frac{1}{j\omega C}$; $Z_3 = j\omega L$; $L = 1 \text{ H}$ e $C = 0,2 \text{ F}$



1º Caso (Apenas fonte de tensão):

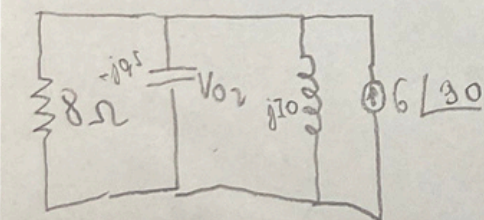
$$Z_1 = 8; Z_2 = \frac{1}{j5 \cdot 0,2} = \frac{1}{j} = -j; Z_3 = j\omega L = j \cdot 1 \cdot 5 = j5$$

$$Z_p = \frac{Z_2 Z_3}{Z_2 + Z_3} = \frac{(-j1) \cdot (j5)}{(-j1) + (j5)} = \frac{-j^2 5}{j4} = \frac{5}{j4} = -j1,25$$



$$V_{01} = \frac{V_1 \cdot Z_p}{Z_1 + Z_p} = \frac{75 \angle 90^\circ \cdot 1,25 \angle 90^\circ}{8 - j1,25} = \frac{93,75 \angle 180^\circ}{8,1 \angle -8,7^\circ} = 11,57 \angle 188,7^\circ$$

2º Caso (Apenas fonte de corrente)



$$Z_1 = 8; Z_2 = \frac{1}{j10 \cdot 0,2} = -j0,5; Z_3 = j10 \cdot 1 = j10$$

$$Z_{1||2} = \frac{8 \cdot (-j0,5)}{8 - j0,5} = \frac{-j4}{8 - j0,5} = \frac{4 \angle 90^\circ}{8,0156 \angle -3,47^\circ} = 0,4930 \angle -86,4237^\circ$$

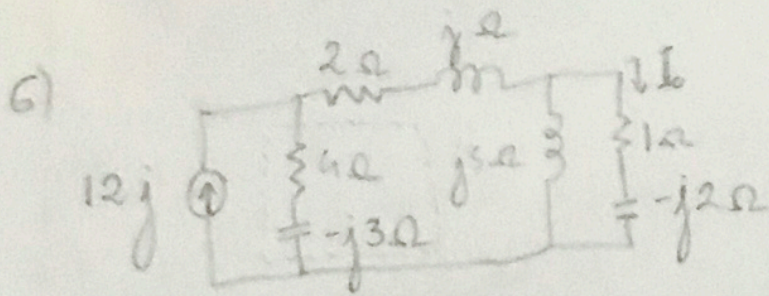
$$Z_{1||2} = 0,4930 \angle -86,4237^\circ = 0,0511 - j0,4970 = Z_{eq} = Z_{1||2} || Z_3 = \frac{0,4930 \angle -86,4237^\circ \cdot 10 \angle 90^\circ}{0,0511 - j0,4970 + j10}$$

$$\Rightarrow \frac{4,930 \angle 3,5763^\circ}{0,0311 + j9,503} = \frac{4,930 \angle 3,5763^\circ}{9,5030 \angle 88,21^\circ} = 0,5250 \angle -84,6337^\circ = 0,03447 - j0,5223$$

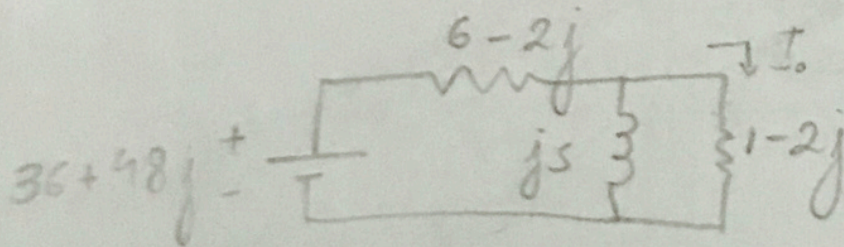
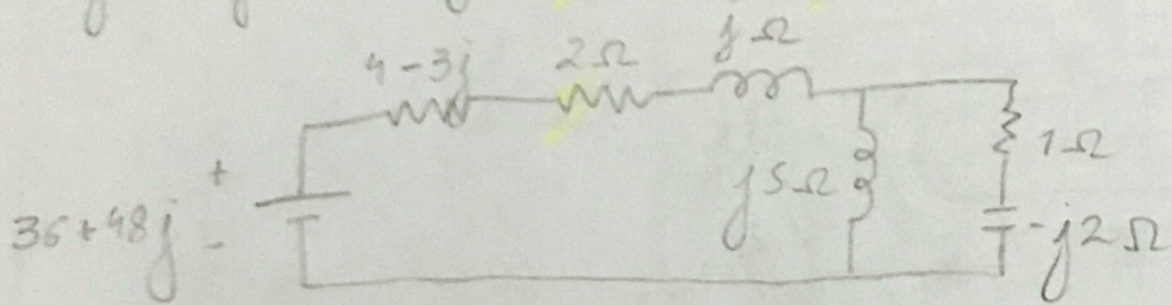
$$V_{02} = Z_{eq} \cdot I = 6 \angle 0^\circ \cdot 0,5250 \angle -84,6337^\circ = 3,15 \angle -84,6337^\circ = 3,15 \cdot \cos(10t - 84,6337^\circ) \text{ V}$$

$$V_{01} = 11,57 \angle 188,7^\circ = 11,57 \angle -171,3^\circ = 11,57 \cos(5t - 171,3^\circ) = 11,57 \cos(5t - 81,13^\circ)$$

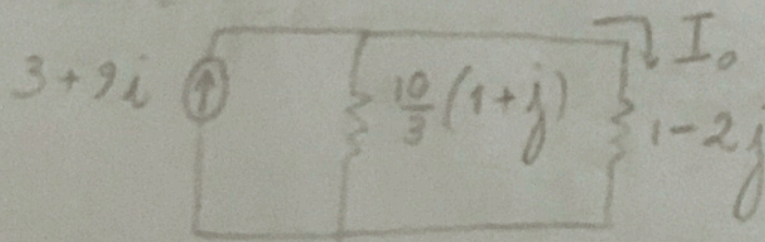
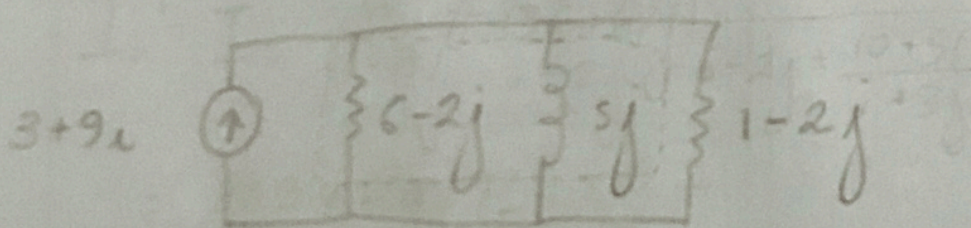
$$V_0 = 11,57 \cos(5t - 81,13^\circ) + 3,15 \cos(10t - 84,6337^\circ) \text{ V}$$



$$V = 12j \cdot (4 - 3j) = 36 + 48j$$



$$I = \frac{36 + 48j}{6 - 2j} = \frac{120 + 360j}{40} = 3 + 9j$$



Divisor de corrente:

$$I_o = (3 + 9j) \frac{\frac{10}{3}(1 + j)}{\frac{10}{3}(1 + j) + (1 - 2j)} = 9,864 \angle 99,462^\circ$$